125101 - Analytical chemistry 1 for engineers

Tutorial 10 – Gran plot in Titration

T10-1. At the equivalence point, in which titration the *y* value depends on the concentration of the analyte? Explain.

A. HCl titration with NaOH for pH

B. HAc titration with NaOH for pH

C. I^- titration with Ag^+ for $p[Ag^+]$

D. Fe^{2+} titration with Ce^{4+} for *E* (potential)

T10-2. Try to develop the equation of Gran plot in a weak acid titrated with a strong base:

$$V_b \times 10^{-pH} = K_a V_e - K_a V_b$$

T10-3. 0.60 g of $Fe(NH_4)_2(SO_4)_2 \cdot 6H_2O$ (M.W 392.13 g/mol) were dissolved in 400 mL of 1.00 M H_2SO_4. The well-stirred solution was titrated with 0.02 M $KMnO_4$, using Pt and calomel reference electrode with a pH meter as a potentiometer. Before use, zero the meter by connecting the two inputs directly to each other and setting the millivolt scale to 0.

Calculate the potentials after addition of different volumes of the titrant of $KMnO_4$:

V = 5.0 mL; V = Ve; V = 17.0 mL

$$MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

$$E^o(Fe^{3+} | Fe^{2+}) = 0.68 \text{ V in } 1.00 \text{ M } H_2SO_4;$$

$$E^o(MnO_4^- | Mn^{2+}) = 1.507 \text{ V};$$

$$E(\text{S.C.E.}) = 0.241 \text{ V}$$

T10-4. which statement(s) is/are incorrect about titration?

a. The greater of difference in reduction potential between analyte and titrant, the sharper will be the end point and the titration error would be smaller.

b. The titration curve shows potential as a function of the volume of added titrant in redox titration.

c. The data set used in Gran plot should be chosen between the equivalence point, and then plot a diagram to calculate the end point of Ve by linear regression analysis.

d. The potentiometer measures the voltage of the cell would not affect the concentrations of reactants in redox titration.